Proper Orthogonal Decomposition and Fourier Analysis on the Energy Release Rate Dynamics of a Gas Turbine Combustor

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The phenomena of combustion dynamics strongly depends on the energy release (flame) dynamics of combustion systems. The temporal Fast Fourier Transform (FFT) and spatial Proper Orthogonal Decomposition (POD) methods were used analyze high speed energy release (chemiluminescene) videos on a natural gas powered gas turbine burner. The considered case has a single dominate acoustical frequency. Both methods provide an insight into the temporal and spatial distribution of energy release rate dynamics associated with the acoustic pressure oscillation. In the FFT analysis, the spatially distributed amplitude and phase angle at the selected acoustic frequency can be secured, and the energy release dynamics (motion) associated with the considered acoustics can be visualized. The trigger mechanism for combustion dynamics was unveiled in the spatial phase angle distribution at the acoustic frequency. In the POD analysis, the first few modes have been found to contain a significant portion of energy associated with the considered dynamics system. The first POD mode contains 19% of energy and its associated temporal and spatial feather is very similar to those calculated by the FFT analysis. The POD mode #2 and #3 contains less energy, 7% and 5% respectively and their temporal behavior is also associated with the acoustical frequency. However, their spatial structures contain higher spatial frequency components compared to that of Mode #1. The unveiled spatial structure by mode #2 and #3 cannot be easily discovered from the FFT amplitude and phase information since it was washed out by the higher energy spatial structure from mode #1. A very significant finding in this research is that the POD can be used for reduced order set of modeling which can capture the energy release dynamics by truncating out the "noise." The POD also represents an effective means of "spatial frequency filtering," based on the POD related energy distribution, which enables the sharpening of the energy release rate spatial structure information.

Nomenclature

a_n	= waveform, time dependent coefficient; n = 1, 2, 3, … M
$A^{(n)}$	= eigenvectors
b_n	= complex number corresponding to amplitude and phase
C	= correlation matrix
f	= the acoustic frequency
i	= $\sqrt{-1}$
M	= total frame number in high speed video
Z	= high speed video data
λ_n	= eigenvalues
v	= video file image
ω	= the angular frequency
$Ø_n$	= POD mode shape; n = 1, 2, 3, … M

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I. Introduction

Combustors of gas turbine engines experience phenomenon referred to as "Screech, Howl and Growl," which is high amplitude pressure oscillations due to instability in the combustion process. The phenomena are also referred to as 'combustion dynamics.' The cause of this instability is believed to be the coupling of the unsteady heat release in the flame with the natural resonance of an acoustic mode in the combustor duct, related to its geometry and configuration. Severe structural vibration and failure of some engine components may result from resonant frequencies and high amplitudes of the combustion Screech phenomenon. Thus uncontrolled combustion dynamics can greatly limit the operability and performance of an engine. The mechanism causing combustion instabilities in gas turbine combustors are not well understood. Although this topic has been widely studied by scientists and engineers, Mani, [2] Yi, et al [3-5] Cai, et al [6], Ichihashi, et al [7] the added geometric complexities and injector configurations of a practical combustor make their dynamical behavior unpredictable using state-of-the-art approaches.

Combustion instabilities are not clearly defined and currently industry design techniques are largely empirical. To improve our understanding of stability properties in such complex systems, encountered in many industrial applications, the flame structure of an atmospheric swirl-stabilized burner incorporating dilution and cooling air holes and fueled with natural gas fuel was systematically investigated for various inlet temperatures, pressure drop, and air–fuel ratios, Mohammad, et al [8-10]. Chemiluminescence imaging with a high-speed CCD camera and a microphone were used to characterize the combustor behavior. Imaging of the combustion field provided an insight on the flame structure. The acoustic emissions of this combustor sector have been surveyed by a microphone for a wide range of equivalence ratios, air inlet temperatures and air pressure drops. Cai [11] showed that the acoustic power spectrum at different overall equivalence ratios are illustrated in Figure 1. The low amplitude acoustic emission was also found around 650 Hz which is corresponding to the ¼ acoustic resonant wave based on the distance and the combustion product temperature from the dome plane to the combustor exit plan. Two high amplitude low-frequency unstable modes (higher overall equivalence ratio conditions with frequency around 400Hz and a lower overall equivalence ratio conditions with an instability frequency around 280 Hz) were measured.

In this paper, the procedure of two different methodologies: temporal Fast Fourier Transform (FFT) and spatial Proper Orthogonal Decomposition (POD), to analyze high speed videos on energy release dynamics (chemiluminescene) are discussed. The case study was conducted on the fuel-lean combustion dynamics with single dominate acoustical frequency around 280 Hz. The temporal and spatial characteristics of the energy release dynamic revealed by both methods are illustrated and compared. Results and discussions on FFT and POD methods on the energy release rate with dual temporal dominant frequencies will be presented in future technical papers.

A common method used to substantially reduce the order of the model is proper orthogonal decomposition (POD). POD techniques are an efficient means to reduce spatially highly complex flow fields by representing them by a small number of spatial modes and their temporal coefficients. This method is an optimal approach in that it will capture the largest amount of the flow energy in the fewest modes of any decomposition of the flow. (Berkooz et al. [12]). POD computes a small number of orthogonal basis functions that contain as much information as needed to represent the original system dynamics. This is accomplished by selecting a set of bases or modes that contain the "most energy" for a particular flow regime. By projecting the data onto the modes, the ensemble can be approximately reconstructed, and thus POD enables analysis of all necessary data in as few modes as possible. POD thus offers a statistical tool for the construction, from the Navier-Stokes equations, of low-dimensional dynamical models for the interaction of essential structures. A common approach referred to as the method of "snapshots" introduced by Sirovich [13] is employed to generate the basis functions of the POD spatial modes from flow-field information obtained using either experiments or numerical simulations. This approach to the modeling of the global wake behavior behind a circular cylinder was effectively employed by Gillies [14]. Details of POD and its application in fluid dynamics and turbulence can be found in the paper by Berkooz et al. [12] and Cohen et al. [15 - 18]. In a recent paper, Siegel et al [19] develop an extension to the POD approach, referred to as 'double proper orthogonal decomposition' (DPOD), in which shift modes have been added to account for the changes in the flow due to transient forcing.

Blue and green color intensity levels of the high-speed CCD camera video data on natural gas fueled combustion represent chemiluminesence of flame dynamics- energy release dynamics. Each pixel of video data in time domain has its own waveform shapes. Identifying waveforms in each pixel of high speed video data let us differentiate data

by color intensity so that a time domain waveform of color intensity is obtained, in other words, frequency behavior and its corresponding region in the combustor can be defined in two dimensional form image. Fast Fourier Transform (FFT) is an ideal method to determine its frequency behavior in image region. These results obtained by the high speed camera correlated with the pressure transducer data. This is one of methods to identify stability characteristic in such complex system.

Proper Orthogonal Decomposition (POD) is a more sophisticated method utilizing the high speed video data that consists of a frame by frame, or snapshot method, to analyze the measured color intensity (heat release rate) properties that are related to the acoustical behavior / combustion dynamics. POD will make it possible for us to create a space and amplitude contour for a specified frequency of interest by defining its mode shape and corresponding time-domain waveform. This will provide detailed analysis of combustor characteristics that are directly related to acoustic phenomena / combustion dynamics. Because we can separate data by frequency and spatially we can visually specify the characteristics of energy release rate in the combustor.

II. Theoretical Notes

Let us consider a function Z(x,y,t) as combustion dynamics captured by high speed video camera. Freedoms in x and y direction represent spatial coordinate, pixel number in video, and t direction is temporal coordinate, frame number.

The Proper Orthogonal Decomposition:

The proper Orthogonal Decomposition is a technique used for separating a signal to spatial and temporal functions. Reconstruction of the original signal via separated functions is also possible. The principal characteristics to be determined by POD are mode shapes, $\emptyset_n(x)$, and its corresponding waveform, $a_n(t)$, of the high speed video.

$$Z(x,y,t) = \overline{Z}(x,y,t) + z(x,y,t) \tag{1}$$

Summation of variable separated form in equation (2) is complete if M approaches to infinity, but a function z(x,y,t) is finite by limiting frame number for our application, therefore exact reconstruction/decomposition is possible.

$$z(x,y,t) \approx \sum_{n=1}^{M} a_n(t)\phi_n(x,y) \tag{2}$$

Expression in equation (2) is called Proper Orthogonal Decomposition (POD). We must note that spatial function is orthonormal to each other given by equation (3) where D is the spatial domain to solve equation (2) efficiently.

$$\int_D \phi_{n_1}(x,y)\phi_{n_2}(x,y)dxdy = \begin{cases} 1 & if\,(n_1 = n_2) \\ 0 & otherwise \end{cases} \tag{3}$$

Spatial function, $\emptyset_n(x)$, can be found by determining eigenfunctions. This technique is called Method of Snapshots referred by Sirovich [2]. This problem can be solved if the data contain equally spaced time intervals. Indeed, our high speed video data fits criteria. Superscripts, i and j, represents frame number.

$$(C)_{ij} = \frac{1}{M}\int_D v^{(i)}(x,y)v^{(j)}(x,y)dxdy \qquad i,j = 1,\ldots,M \tag{4}$$

M is total frame number of data, and M x M matrix C is constructed by equation (4) using frame by frame high speed video. Relationship of eigenfunctions and eigenvectors is expressed in Equation (5). The magnitude of corresponding eigenvalues to modes represents energy of its POD modes.

$$CA^{(n)} = \lambda_n A^{(n)} \qquad n = 1,\ldots,M \tag{5}$$

Substituting the eigenfunctions defined from equation (5) into equation (6), we are able to define spatial function, $\emptyset_n(x)$, which is POD mode shape. Note that eigenfunctions are linear combination of high speed video data.

$$\phi_n(x,y) = \sum_{k=1}^{M} A_k^{(n)} v^{(k)}(x,y) \qquad n = 1,\ldots,M \tag{6}$$

Waveform, $a_n(t)$ which is corresponding with POD nth mode shape, is expressed by equation (7). Orthonomality in spatial function, $\emptyset_n(x)$, in mind, temporal coefficient, $a_n(t)$, that only depends on $\emptyset_n(x)$ is solved.

$$a_n(t) = \int_D z(x,t)\phi_n(x)dx \tag{7}$$

The Fourier Transform:

DC component is taken away as shown in equation (1). AC component in Fourier Transform is expressed in equation (8).

$$z(x,y,t) \approx \sum_{n=1}^{M} b_n(x,y) e^{i\omega_n t} \tag{8}$$

$$\int_D e^{i\omega_n t} e^{i\omega_m t} dt = \begin{cases} 1 & if\,(n=m) \\ 0 & otherwise \end{cases} \tag{9}$$

Where $b_n(x,y)$ is a complex number representing the amplitude and phase angle associated with a specific frequency at a specific spatial location. In Fourier Transformation, temporal function is orthonormal to each other. The form resemble to the equation (3) which expressed orthonormality in spatial function.

III. The Experimental Approach

Experimental rig configuration:

The combustor used in the experiments described in this paper is shown in Figure 2. It is a segment of a single-annular combustor from a prototype gas turbine engine. The ducting containing the preheated airflow upstream of the combustor is schematically shown in Figure 2. An AKG D112 microphone was employed in the test cell to monitor the combustion noise radiating from the exhaust termination of the combustor. The microphone is capable of sensing 20Hz to 17,000Hz At one side of the burner, a quartz window was mounted for flow visualizations. At the other side of the burner, a stainless steel plate, having the same size as the quartz window, was installed with four PCB dynamic pressure transducers. A Phantom V7.3 high-speed CCD video camera was stationed facing to the combustor window to capture the flame dynamics. Figure 4 shows a flame structure image captured by the high speed video camera when the burner is operated at very high equivalence ratio.

High Speed Video Processing:

Video from the high speed video camera is stored in AVI file format with 800 x 600 resolutions. Video data requires appropriate conversion from AVI file in order to apply the in-house developed POD or FFT tools which is based on National Instrument LabView Version 7.1 software. First, the AVI file format is converted into grayscale image, depending on the user's choice on the color (RGB) selection, then the pixels of an image are extracted to 2D array of 32bits (float) number. The original 800 x 600 pixels are reduced down to 200 x 150 cells by taking average of 4 x 4 pixels. The current approach has adequate cells to capture the characteristic of combustion dynamics with significantly less computational sources requirements. This file conversion loop continues until the last frame of

AVI video file and 2D array that is converted during the loop is saved in memory, resulting 3D array that has depth for total frame length, showing in Figure 3. The 3D array created here is the function Z(x,y,t) expressed in equation (1) and can be processed by the FFT and POD software.

IV. Results and Discussions

To investigate the connections between acoustic emissions and flame dynamics, the high speed video data for flame energy release rate and acoustic emissions were conducted in a prototype gas turbine combustor sector. A test case, with combustion dynamics at 280Hz, was selected to illustrate the different capabilities of the two considered (POD and FFT) analysis methods. Radiating acoustic wave from the experimental combustor recorded by a microphone is dominated by a single frequency band at 280 Hz as shown in Figure 4. A single frame of high speed video is shown in Figure 5. The time-average mean and rms chemiluminescence intensity from 3000 video frames are shown in Figures 7 and 8. The peak of flame dynamics occurs at the swirling flames inside the combustor dome (red color in Figure 8).

FFT Analysis:

The temporal FFT spectrum based on the total integrated intensity (sum of all pixel intensity) of each frame was first conducted. The resulted energy release amplitude spectrum, as shown in Figure 8, is similar to the acoustic spectrum in Figure 4 and is dominated by a single frequency band around 279.5 Hz. This indicates the acoustic emissions and energy release dynamics are closely related. The temporal FFT, which is based on the finite number of video frames (3000), has frequency resolution of 2.2 Hz. The spatially distributed FFT amplitude at three neighboring frequencies around 280 Hz (277.3, 279.5 and 281.7) are shown in Figure 9. The amplitude levels at 277.3 and 281.7 Hz are significantly lower than those at 279.5 Hz indicating extremely narrow band flame oscillations at 279.5 Hz.

The spatially distributed phase angle (radian) contours associated with 279.5 Hz are shown in Figure 10. From Figures 9 and 10, the 279.5 Hz flame dynamics are concentrated in three amplitude local maximums. Two local maximum spots, (45,100) and (55,25) are near the swirler exit with much higher amplitudes than the third local maximum spot, (80,65), at the centerline downstream. Also, the two spots with high amplitude coincidence of the highest flame dynamics spots of the r.m.s. and mean amplitude contours shown in Figures 7 and 8 indicating the acoustic energy source is the flame front oscillations. The majority of phase angles at the high amplitude regions have value close to 3.14 or -3.14 radian indicating the majority of energy release from the flame are in phase. However, there is local maximum phase angle around 0 radian located at the lower swirler exit area, (55,25); indicating some energy release rate from the flame front is out of phase with the main flame dynamics. Actually, two local extreme spots , (55,25) and (45,100) near the swirler exit has phase angle of 180 degree which provides clues that a small circumferential energy release dynamics may be the triggers for the main combustion dynamics.

Based on the amplitude and phase contours (Figures 9 and 10), we can reconstruct the motions of energy release rate associated with 279.5 Hz acoustics. Figure 11 illustrates one complete cycle of 279.5 Hz dynamics: 12 plots with 30 degree phase angle increment. The physical time step associated with 30 degree phase angle is about 0.3 ms (1/279.5*30/360 second). The illustrated motion is essentially the averaged (279.5 Hz) cyclic motions based on total number of cycles (about 120 cycles in the current study) on the video recordings. The variations among cycles tend to be washed out in Figure 11. As discussed before, majority of energy release in the whole combustor is in phase: dominated by the same color, either blue (negative values) or brown (positive values), in the individual phase angle plot. However, there are mixed colors (positive and negative values) in the plots with phase angles of 90, 120, 270 and 300. The amplitude levels in those four plots are much lesser than that in other phase angles. Furthermore, the spatial structures are similar among these four plots but the color is reversed between plots with 180 phase angle difference. The above observations support physical phenomena of a “small” circumferential energy release dynamics co-exists with a “big” in-phase energy release dynamics at 279.5 Hz. The wavelength of 279.5 Hz acoustics is significantly longer than the current spatial domain captured by high speed video. If positive feedback exists between acoustics and energy release (thermal acoustic instability), the resulted high amplitude energy release should be in-phase. Again, circumferential energy release dynamics may be the triggers for the main combustion dynamics is concluded.

Figure 12: shows roughly one cycle (279.5 Hz) of original video on dynamics part of energy release rate. Plots are increased by two video frames which is corresponding to 30.5 phase angle at 279.5 Hz. It is extremely hard to visualize the spatial structure associated with 279.5 Hz cyclic behavior. Motion plots, Figure 11, based on FFT amplitudes and phase angles at 279.5 Hz essentially act as the "best" narrow band filter at 279.5 Hz to isolate the motion associated with this frequency.

POD Analysis:

This technique (POD) is optimized so that it will analyze the largest amount of system energy with the least (spatial) modes of any system dynamics decomposition, which is suitable method for describing/modeling a complicated energy release structure. Therefore, using POD will enable us to create a few (spatial) mode shapes which contain the majority of energy of the analyzed dynamic system and the reduced dynamic models can be more easily established. Temporal behavior associated with each mode may contain different frequencies and phase angles, therefore, be able to pin point which the (spatial) mode shapes associate with frequency of interests. In this effort we will show that POD is an effective spatial filtering tool for the application examined.

The first three spatial mode shapes and its temporal spectra are plotted in Figures 13-15. All modes shown here have different shapes but their corresponding time-domain waveforms have same frequency (279.5 Hz). Samples of time domain waveforms are illustrated in Figure 16, and it is evident that small phase differences exist between modes even though they oscillate in same frequency. Mode energies are calculated while solving equation (5) which is the corresponding eigenvalues to modes. Accumulated energy up to the first 125 modes is shown in Figure 17. In the current analysis, 3,000 spatial modes and associated energy have been resolved since 3,000 video frames are considered in the POD analysis. The first 125 modes contains 85% of energy of total 3,000 modes, and the first three modes contains 19%, 7% and 5% of energy, respectively. The evident of POD analysis can use the "minimum" number of spatial modes to represent the dynamic behavior is validated.

In the POD analysis, spatial modes cannot have same frequency without phase shift; otherwise, these spatial modes can be combined into a single spatial mode with higher energy. In the FFT analysis, the phase angles are depending on the spatial locations; different phase angles at different locations. However, in the POD analysis, each spatial mode has one temporal waveform so the whole mode oscillates as a unit modulated with associated temporal waveform. If the waveform is dominated by a single frequency, positively valued regions and negative valued regions in the spatial mode shape will oscillate out-of-phase.

The POD mode #1 structure is very similar to the amplitude contour plots by FFT (Figure 9). Three local maximums and minimums, (55,25), (45,100) and (80,65), are in the same locations as the local maximums in the amplitude plots of FFT. Maximums (positive values) and minimums (negative values) spots in the POD, as discussed in the paragraph, has phase difference of 180 degree which is also calculated by the phase angles by FFT in Figure 10. It can be concluded that the POD Mod #1 has extremely similar dynamics, both temporal and spatial behavior, as those isolated by the FFT filtered at 279.5 Hz. The POD analysis also predicts the 18% of dynamic energy is associated with this mode.

POD mode #1 and its corresponding time domain waveform is reconstructed and shown in Figure 18. Plots are increased by two video frames which is corresponding to 30.5 phase angle at 279.5 Hz and the 12 plots present almost a complete cycle. Since the time domain is dominated by a single frequency, the locations of local maximums and minimums are not changed with time or phase angle. This is significantly different from the reconstruction generated by the FFT analysis. Figure 11 shows that the local extremes are moving with phase angle or time since the local amplitude is modulated by difference phase angle and higher spatial frequency structure exists at different phase angle plots. The "continuous" motion and higher frequency structure cannot be captured by a single POD mode with associated time domain waveform. The supplemental information will be provided by the higher POD modes which contains less individual mode energy.

The POD mode #2 and #3 contains more local extremes with shorter separation distance (higher spatial frequency) compared to the POD mode #1. The energy associated with #1 and #2 modes are 9% and 5%, respectively which is significantly less than 18% of Mode #1. Local extremes contained in the Mode #1 and #2 are shown up at FFT reconstructed plots (Figure 11) at different phase angles but theses extremes has less amplitude compared to the

peaks identified by Mode #1. This result also verifies the statement that POD is suitable tool for modeling complicated phenomena such as fluid motion and/or combustion dynamics because it capture/describe high dimensional data by minimum information, at the same time, researchers find characteristics of dynamics phenomena.

V. Conclusions

In this effort, we have demonstrated two methods, FFT and POD, for analysis of energy release dynamics experimental data based on high speed video. Each method has its distinct advantages and they **complement one another**. FFT provides the essential temporal characteristic of the resonant frequencies and their relative strengths. POD then can provide additional information concerning the **spatial behavior of the coherent structures** at these resonant frequencies. This is achieved by the POD which provides several modes at the same frequency having phase differences. A very significant finding in our research is that the POD allows us to use a reduced order set of modes to recreate the combustion dynamics by truncating out the "noise" thereby yielding a pristine perspective. This feature of POD represents an **effective means of "spatial filtering"** which enables the sharpening of the video data.

Possible future application with POD:

Utilizing Proper Orthogonal Decomposition method will make it possible for us to create a space and amplitude contour for a specified frequency of interest by defining its mode shape and corresponding waveform, in other words, be able to obtain special and temporal information separately. Earlier study showed that employing this method is useful to understand dynamical behavior in Combustion but required a high-speed CCD camera on a combustor employing a window for the purpose of view in combustion dynamics, therefore, real time POD on current combustor using high-speed CCD camera or application for a real combustor without a window are impossible. To perform POD analysis, high-speed CCD camera is an essential tool, but it is not an acceptable tool for real time POD, instead, pressure transducers are widely available and be able to mount on combustor as well as quick data analysis is commonly done. If each pixel of high-speed video is replaced with pressure transducers, then POD obtains the same result as the POD performed from high-speed video, but it is again nearly impossible to place significant amount of transducers on a combustor.

The temporal portion of POD, corresponding waveform, directly correlates with pressure information, therefore if the pressure information is available, then reconstruction of POD mode becomes possible, in other words, pressure information could definite POD mode shape by matching up its corresponding temporal portion of POD mode data sets. This is time and cost effective way to estimate POD mode which leads to potential real time POD application. Note that dominant modes in combustor are lower modes, therefore, interests in reconstruction of POD modes are limited in small number, and this is one of reasons why small numbers of pressure transducers are only necessary to perform real time POD. Obtaining enough POD data sets from experimental or lab environment prior to real time POD mode analysis on the particular combustor is crucial. Without priory obtained P.O.D data sets available, it is impossible to estimate POD modes from limited pressure transducer information.

If real time POD is achievable then closed loop health monitoring system, addressing problems by mode shape, etc, is possible under relatively quickly and less effort. This is just one of many possibilities by performing real time POD.

POD is known as suitable tool for not only separating time and space but also low dimensional modeling. Results from POD and FFT methods are directly compared and both methods equally defined its characteristic, dominant frequency (frequencies) as the same from subjected combustion dynamic data. By obtaining POD mode shape and its corresponding waveform, the combustor can be easily modeled. Energies of corresponding POD modes are also defined during process of POD calculation. How many modes required to model combustion dynamic behavior is simply defined by knowing mode energies distribution, which makes complex combustion modeling easier.

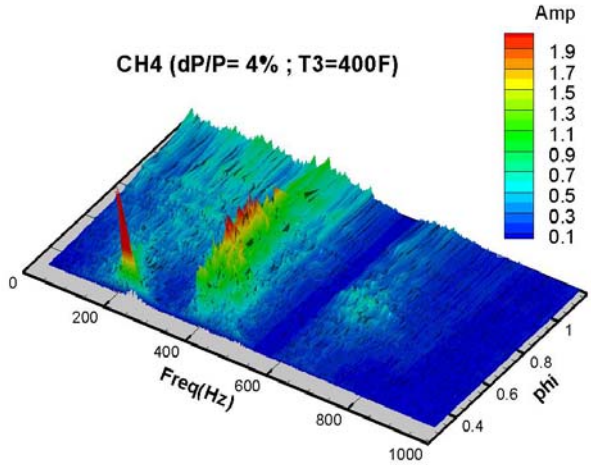

Figure 1 Acoustic Spectrum of the Combustor Sector

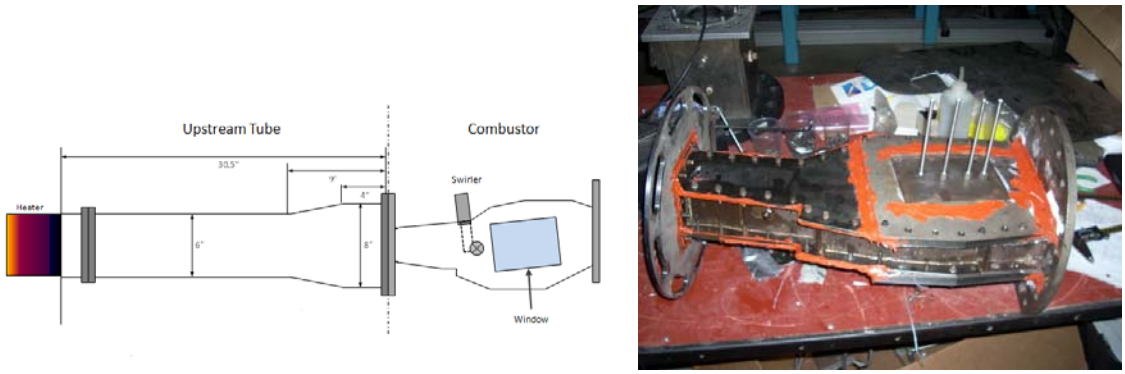

Figure 2 Combustor Hardware and Test Arrangement

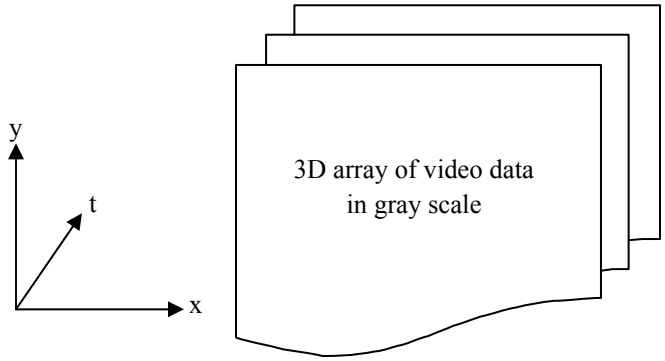

Figure 3: 3D array, Z(x,y,t) Created by LabView

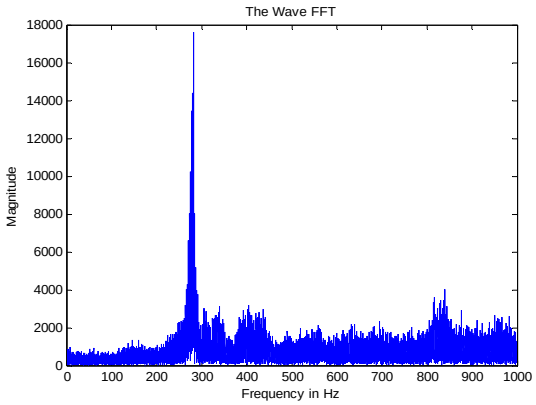

Figure 4: Acoustic Emission Spectrum

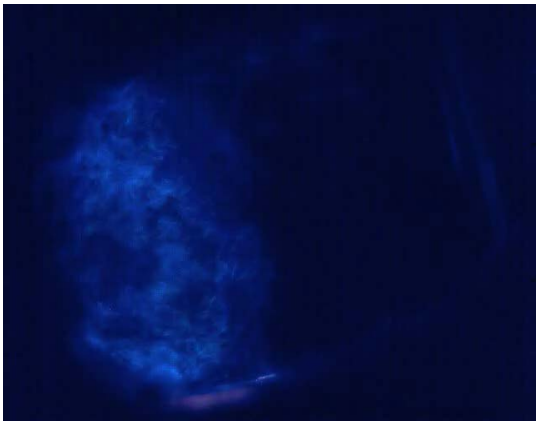

Figure 5: A Frame of High Speed Video

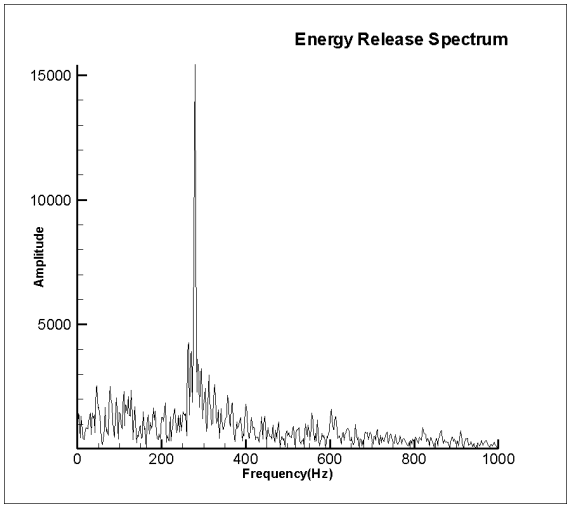

Figure 4: Energy Release Spectrum

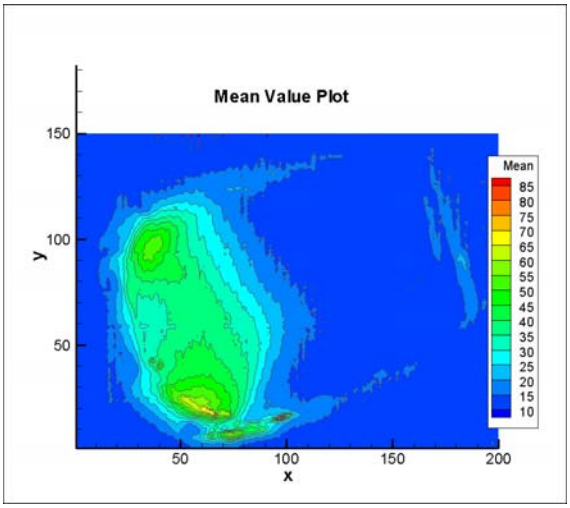

Figure 7: Mean of Energy Release

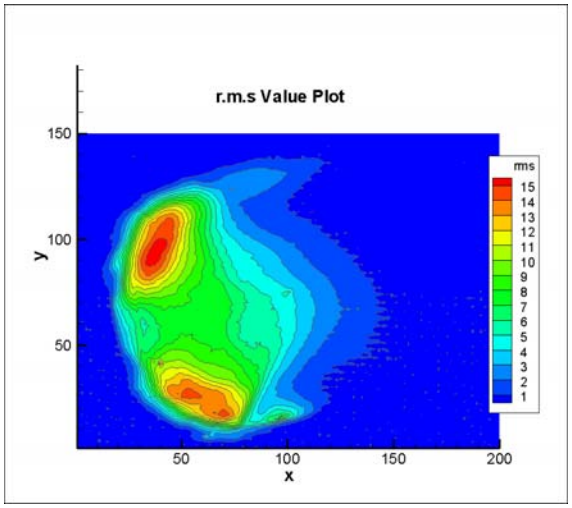

Figure 8: r.m.s. of Energy Release

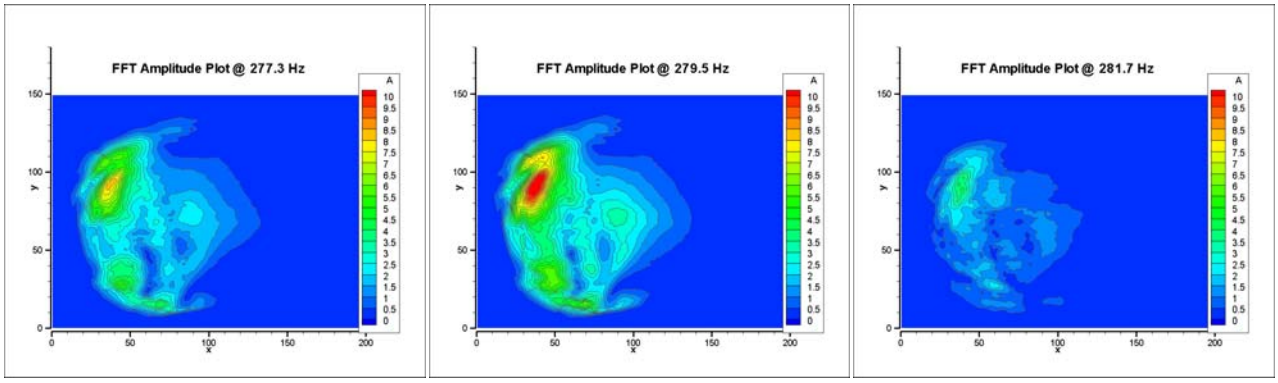

Figure 9: Amplitude Contours of Energy Release at 277.3, 279.5, and 281.7 Hz

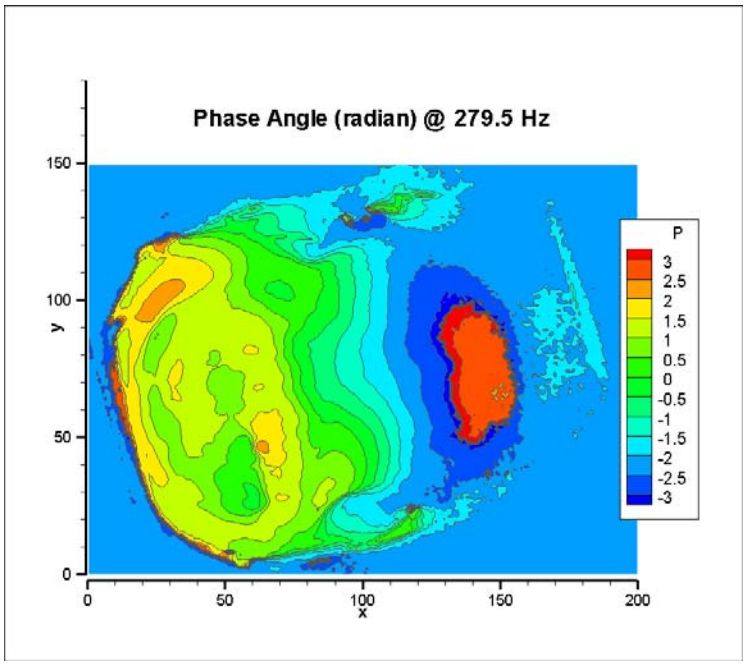

Figure 10: Phases Angle (radian) Contours at 279.5 Hz

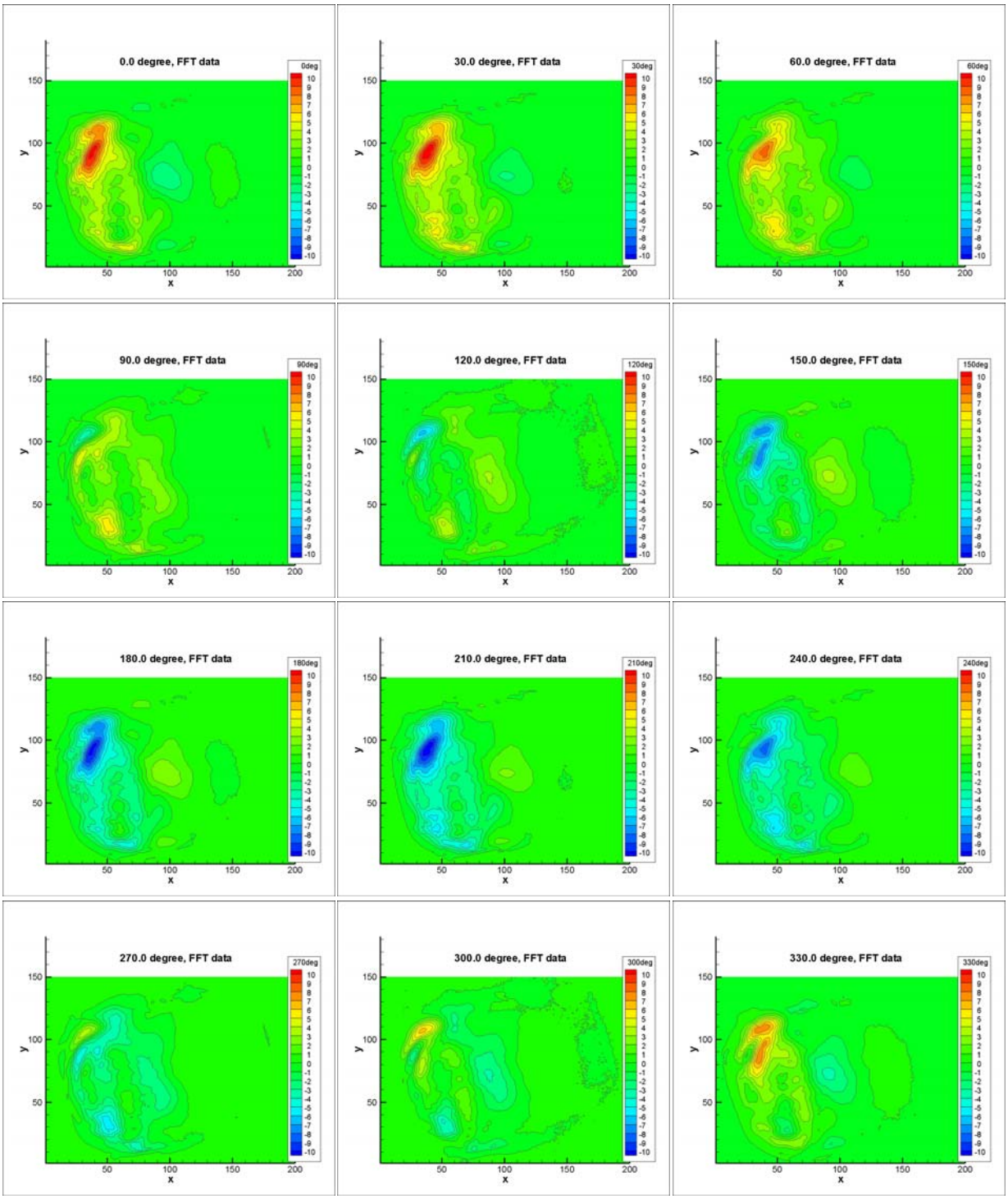

Figure 11: One Cycle of Energy Release Motion at 279.5 Hz (12 plots with 30° increments)

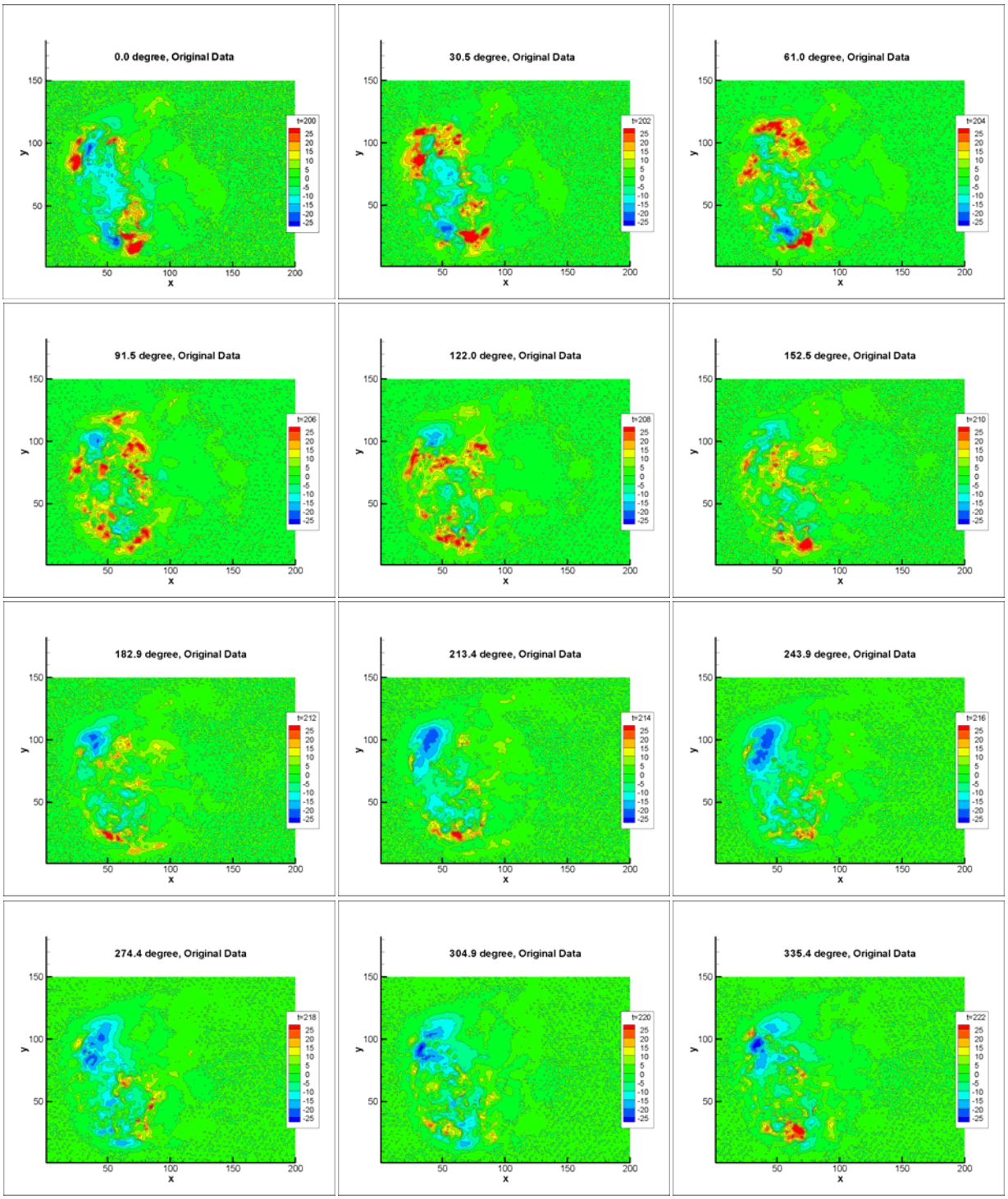

Figure 12: (Approximate) One 279.5 Hz Cycle of Original Video on Fluctuating Energy Release Component (Start with Frame 200 with Frame Increment of 2 or Phase Angle Increment of 30.5 degree)

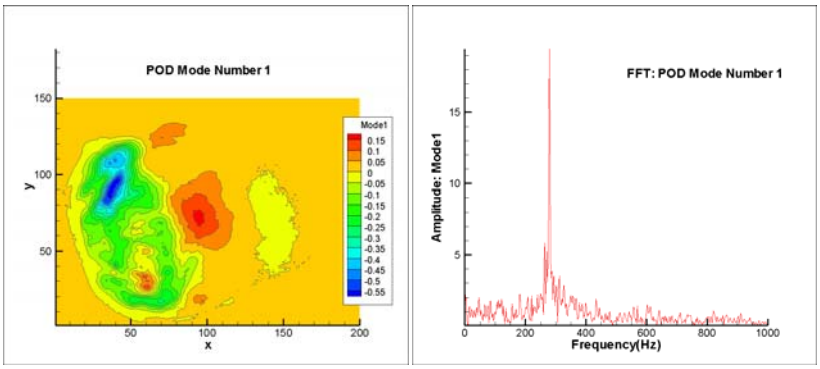

Figure 13: POD Mode #1 Contours and Associated Temporal Spectrum

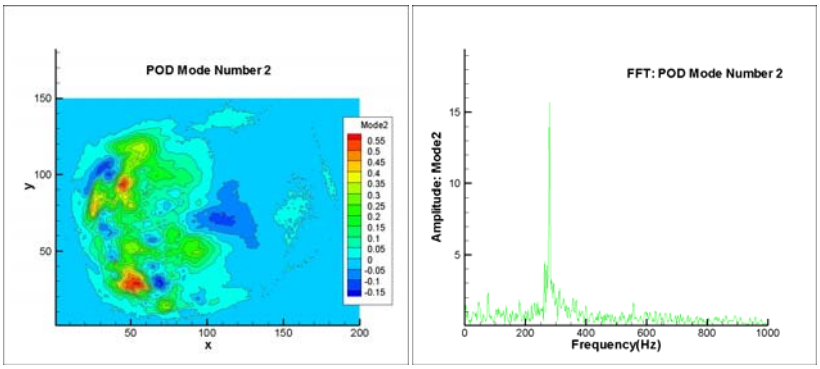

Figure 14: POD Mode #2 Contours and Associated Temporal Spectrum

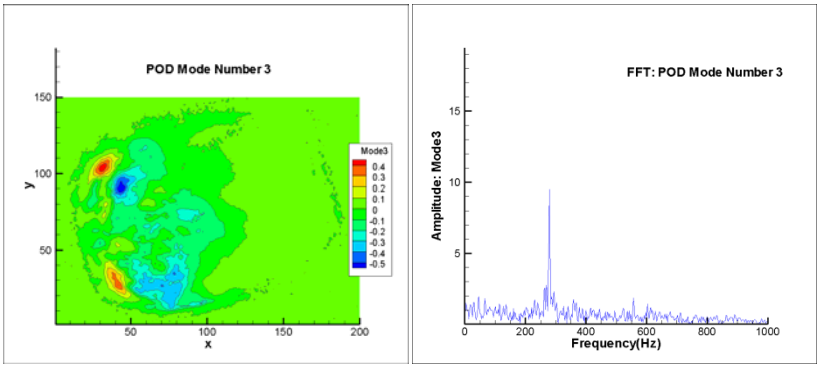

Figure 15: POD Mode #3 Contours and Associated Temporal Spectrum

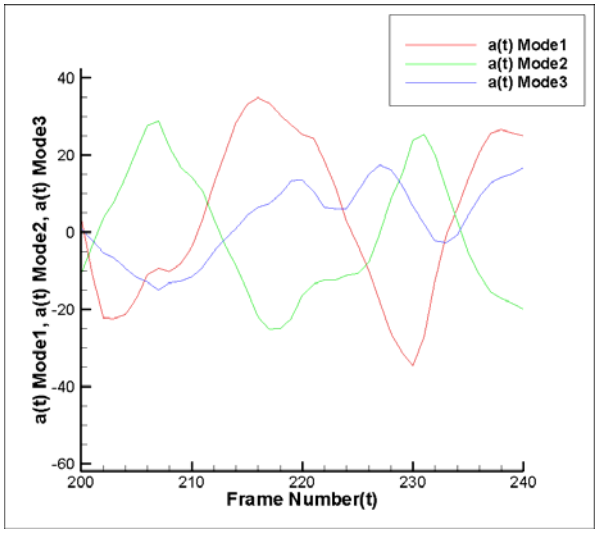

Figure 16 POD Temporal Coefficients as a Function of Time

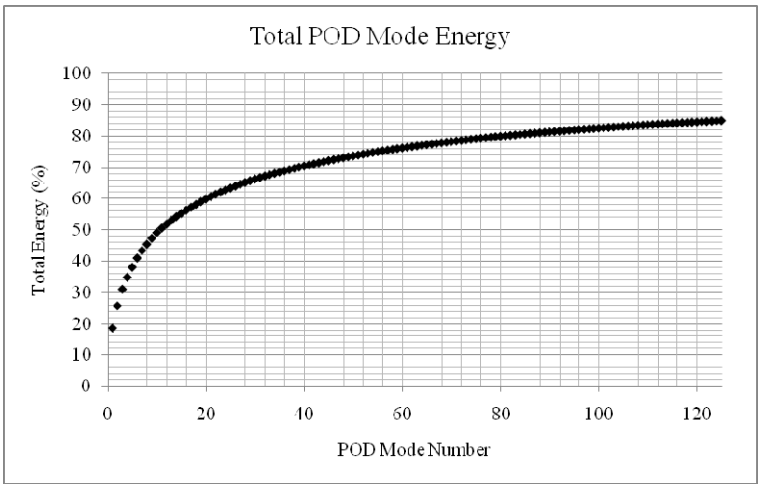

Figure 17 POD Accumulated Mode Energy

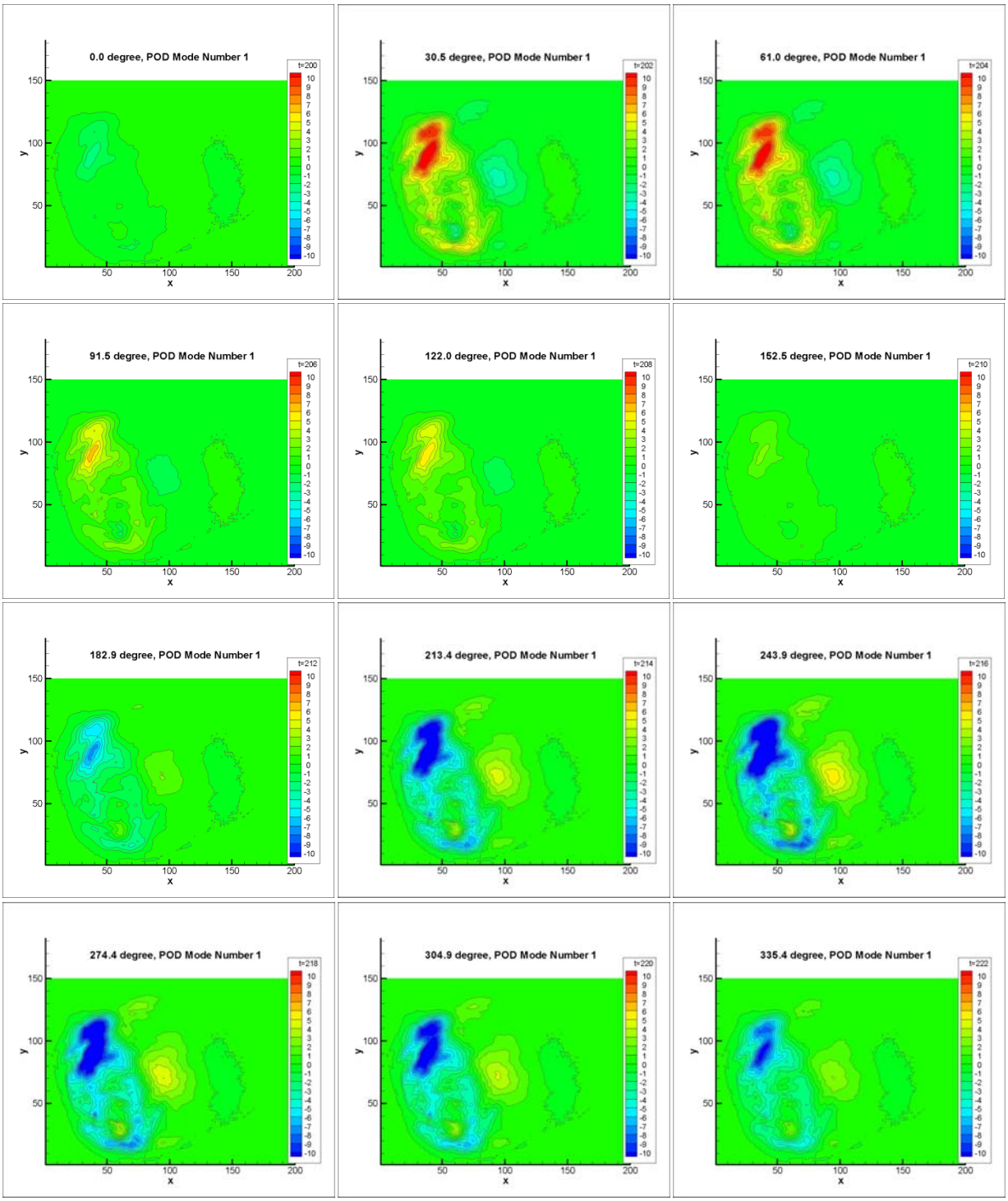

Figure 16: (Approximate) One 279.5 Hz Cycle of POD Mode #1 Reconstruction (Start with Frame 200 with Frame Increment of 2 or Phase Angle Increment of 30.5 degree)

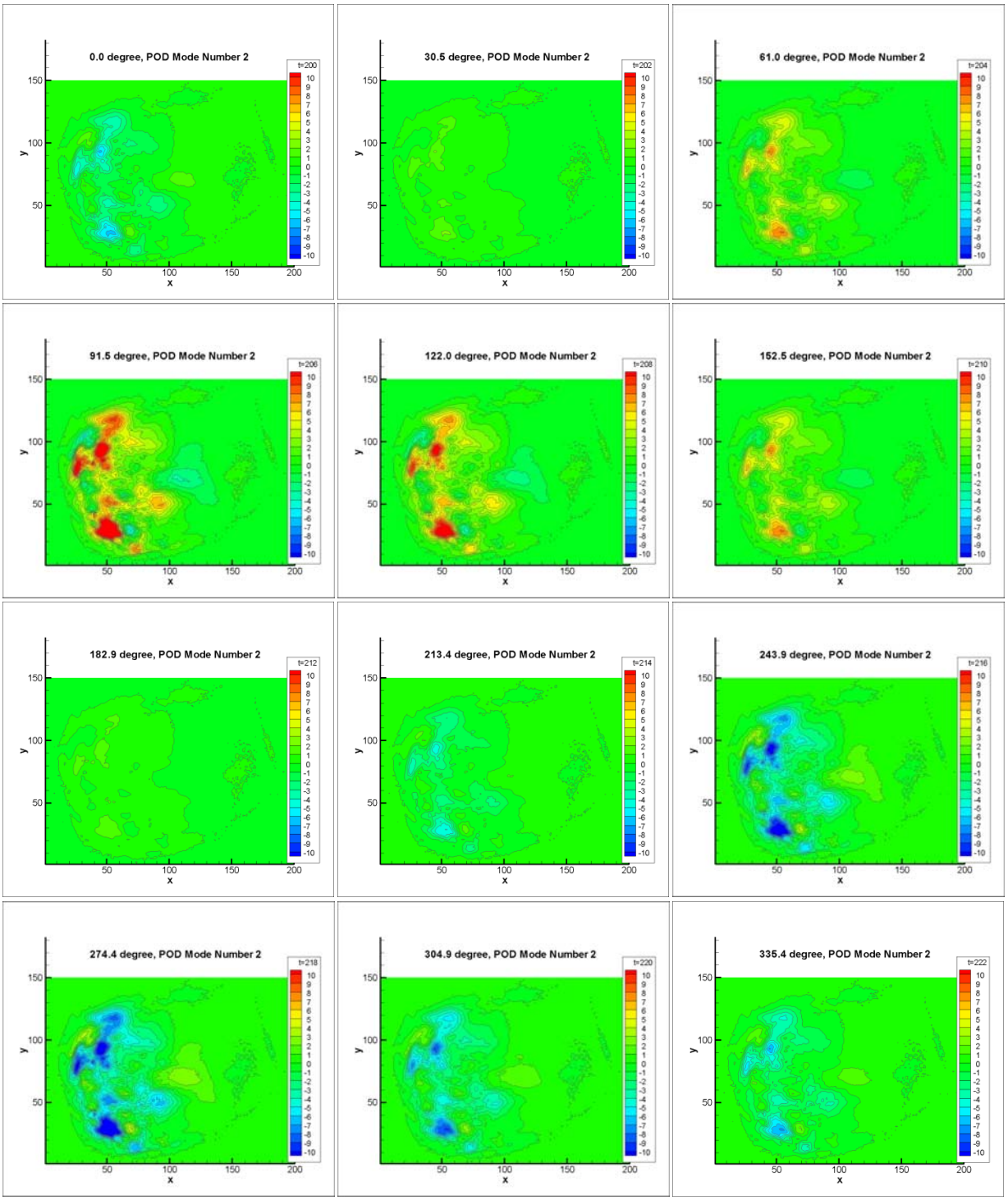

Figure 17: (Approximate) One 279.5 Hz Cycle of POD Mode #2 Reconstruction (Start with Frame 200 with Frame Increment of 2 or Phase Angle Increment of 30.5 degree)

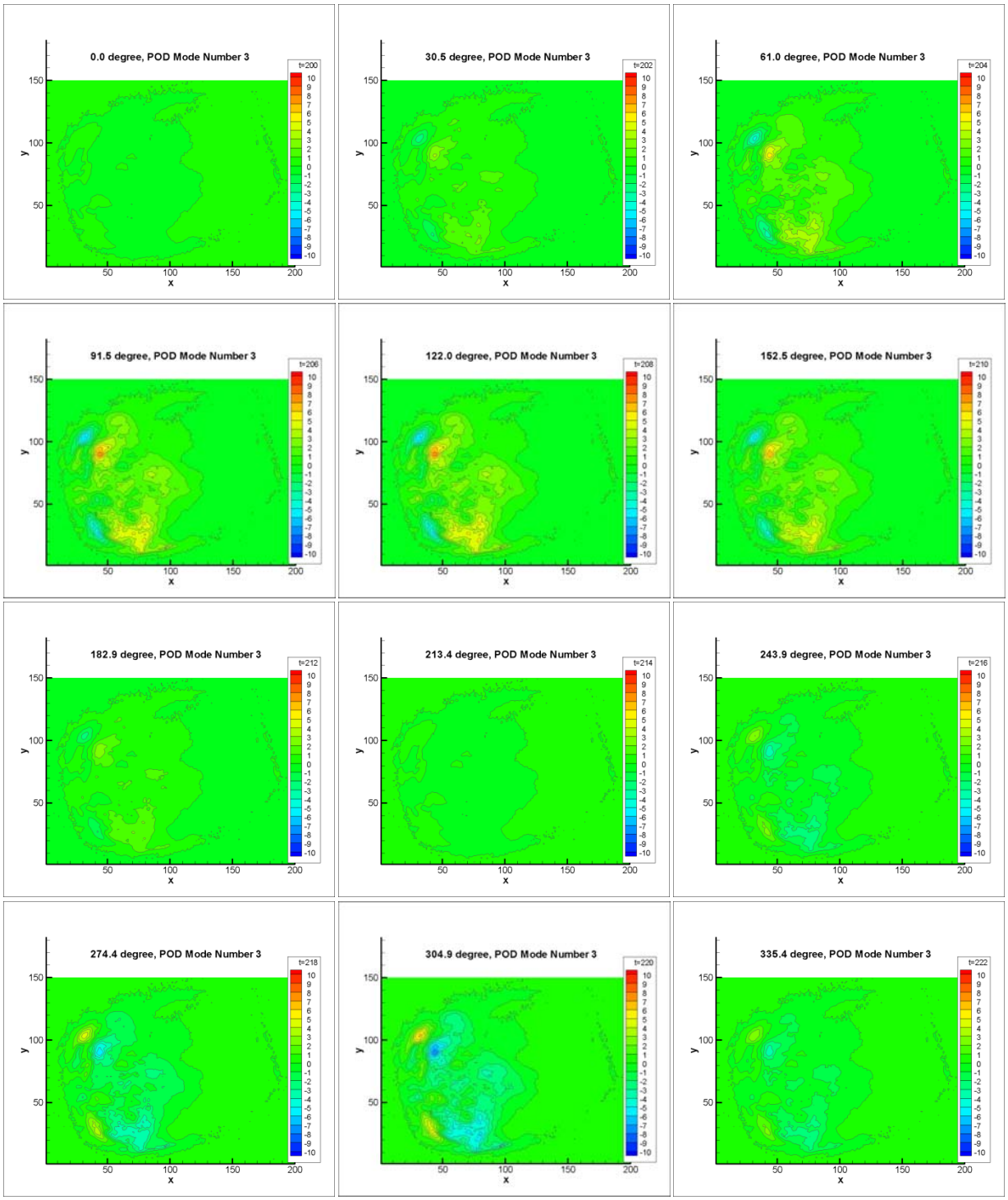

Figure 18: (Approximate) One 279.5 Hz Cycle of POD Mode #2 Reconstruction (Start with Frame 200 with Frame Increment of 2 or Phase Angle Increment of 30.5 degree)

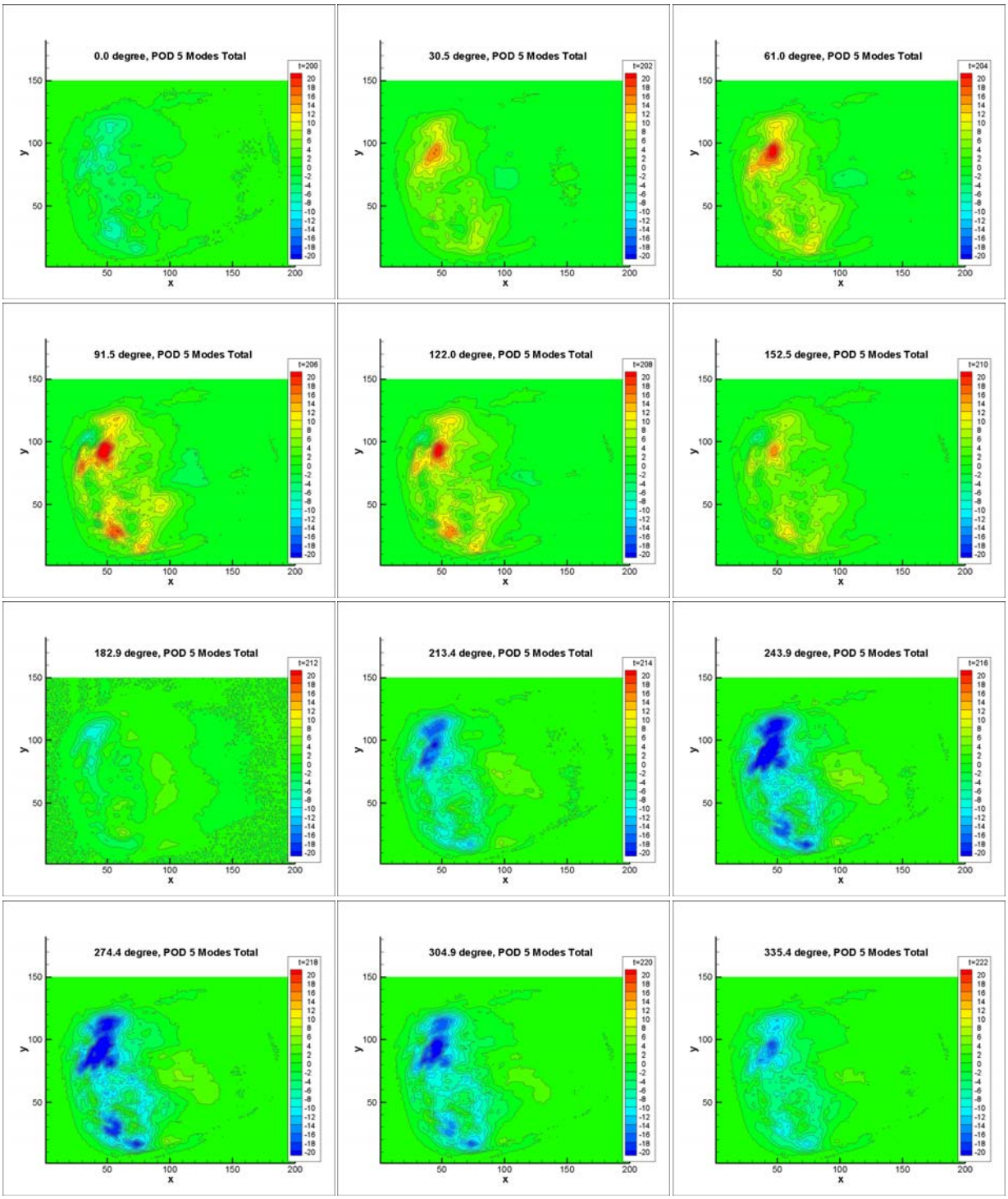

Figure 19: (Approximate) One 279.5 Hz Cycle of Sum of First Five POD Reconstruction (Start with Frame 200 with Frame Increment of 2 or Phase Angle Increment of 30.5 degree)

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